

# Jaechul Roh

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## EDUCATION

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### University of Massachusetts Amherst

*Ph.D. Candidate in Computer Science*

Advisor: Prof. [Amir Houmansadr](#)

GPA: 4.0/4.0

September 2023 – May 2028 (expected)

Amherst, Massachusetts, USA

### University of Massachusetts Amherst

*M.S. in Computer Science*

Advisor: Prof. [Amir Houmansadr](#)

GPA: 4.0/4.0

September 2023 – May 2026

Amherst, Massachusetts, USA

### Hong Kong University of Science and Technology

*B.Eng. in Computer Engineering, School of Engineering*

Final Year Thesis Advisor: Prof. [Jun Zhang](#)

\*2 years of Compulsory Korean Military Duty

September 2017 – May 2023

Clear Water Bay, Hong Kong

## RESEARCH INTERESTS

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My research focuses on the **Safety & Security** of AI models and agents. Under the supervision of Prof. Amir Houmansadr, I study the trustworthiness of multimodal generative and interactive models across **audio**, **text**, and **vision** domains. I currently collaborate closely with Prof. Eugene Bagdasarian, as well as researchers at Google DeepMind and Qualcomm.

## PUBLICATIONS

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### Under Review

#### 1. Codec-Robust Attacks on Audio LLMs

Jaechul Roh, Jean-Philippe Monteuis, Jonathan Petit, Amir Houmansadr

*Preprint at arXiv (Under Review)*

[\[paper\]](#) [\[code\]](#)

#### 2. Benign Fine-Tuning Breaks Safety Alignment in Audio LLMs

Jaechul Roh, Amir Houmansadr

*Preprint at arXiv (Under Review)*

[\[paper\]](#)

#### 3. Bob's Confetti: Phonetic Memorization Attacks in Music and Video Generation

Jaechul Roh, Zachary Novack, Yuefeng Peng, Niloofer Miresghallah, Taylor Berg-Kirkpatrick, Amir Houmansadr

*Preprint at arXiv (Under Review)*

[\[paper\]](#) [\[demo page\]](#) [\[video\]](#)

#### 4. SPILLage: Agentic Oversharing on the Web

Jaechul Roh, Eugene Bagdasarian, Hamed Haddadi, Ali Shain Shamsabadi

*Preprint at arXiv (Under Review)*

[\[paper\]](#) [\[code\]](#) [\[article\]](#)

#### 5. OverThink: Slowdown Attacks on Reasoning LLMs

Abhinav Kumar, Jaechul Roh, Ali Naseh, Marzna Karpinska, Mohit Iyyer, Amir Houmansadr, and Eugene Bagdasarian

*Preprint at arXiv (Under Review)*

[\[paper\]](#) [\[code\]](#)

#### 6. Throttling Web Agents Using Reasoning Gates

Abhinav Kumar, Jaechul Roh, Ali Naseh, Amir Houmansadr, Eugene Bagdasarian

*Preprint at arXiv (Under Review)*

[\[paper\]](#) [\[code\]](#) [\[demo page\]](#)

### Conference

#### 1. Multilingual and Multi-Accent Jailbreaking of Audio LLMs

Jaechul Roh, Virat Shejwalkar, Amir Houmansadr

COLM 2025

[\[paper\]](#) [\[code\]](#)

2. **Backdoor**ing Bias ( $B^2$ ) into Stable Diffusion Models

Ali Naseh, **Jaechul Roh**, Eugene Bagdasarian, Amir Houmansadr

*USENIX Security '25*

[\[paper\]](#) [\[code\]](#)

3. **OSLO**: One-Shot Label-Only Membership Inference Attacks

Yuefeng Peng, **Jaechul Roh**, Subhransu Maji, Amir Houmansadr

*NeurIPS 2024*

[\[paper\]](#)

4. **Memory Triggers**: Unveiling Memorization in Text-To-Image Generative Models through Word-Level Duplication

Ali Naseh, **Jaechul Roh**, Amir Houmansadr

*The 5th AAAI Workshop on Privacy-Preserving Artificial Intelligence*

[\[paper\]](#)

5. **Robust Smart Home Face Recognition** under Starving Federated Data

**Jaechul Roh**, Yajun Fang

*IEEE International Conference on Universal Village (IEEE UV2022)*

*Oral Presentation*

[\[paper\]](#)[\[code\]](#)[\[slides\]](#)[\[video\]](#)

6. **MSDT**: Masked Language Model Scoring Defense in Text Domain

**Jaechul Roh**, Minhao Cheng, Yajun Fang

*IEEE International Conference on Universal Village (IEEE UV2022)*

*Oral Presentation*

[\[paper\]](#)[\[code\]](#)[\[slides\]](#)[\[video\]](#)

7. **Impact of Adversarial Training on the Robustness of Deep Neural Networks**

**Jaechul Roh**

*2022 IEEE 5th International Conference on Information Systems and Computer Aided Education (ICISCAE)*

[\[paper\]](#)[\[code\]](#)

## Preprints

1. **World-To-Image: Grounding Text-to-Image Generation with Agent-Driven World Knowledge**

Moohyun Son, Jintaek Oh, Sunbin Mun, **Jaechul Roh**, Sehyun Choi

*Preprint at arXiv (Under Review)*

[\[paper\]](#) [\[code\]](#)

2. **Chain-of-Code Collapse: Reasoning Failures in LLMs via Adversarial Prompting in Code Generation**

**Jaechul Roh**, Varun Gandhi, Shivani Anilkumar, Arin Garg

*Preprint at arXiv*

[\[paper\]](#) [\[code\]](#)

3. **R1dacted: Investigating Local Censorship in DeepSeek's R1 Language Model**

Ali Naseh, Harsh Chaudhari, **Jaechul Roh**, Mingshi Wu, Alina Oprea, Amir Houmansadr

*Preprint at arXiv*

[\[paper\]](#)

4. **FameBias: Embedding Manipulation Bias Attack in Text-to-Image Models**

**Jaechul Roh**\*, Andrew Yuan\*, Jinsong Mao\*

*Preprint at arXiv (Equal Contribution\*)*

[\[paper\]](#)

5. **Understanding (Un)Intended Memorization in Text-to-Image Generative Models**

Ali Naseh, **Jaechul Roh**, Amir Houmansadr

*Preprint at arXiv*

[\[paper\]](#)

## Magazine Article

1. **Can You Trust What AI Hears (and Says)?**

**Jaechul Roh**

*XRDS: Crossroads, The ACM Magazine for Students 31 (4), 34-39*

## HONORS & AWARDS

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### 2026 Google PhD Fellowship

April 2026

*UMass Amherst Nominee*

- Selected as departmental representative candidate for the fellowship

### 2026 Apple Scholars in AI/ML PhD Fellowship

September 2025

*UMass Amherst Nominee*

- Selected as departmental representative candidate for the fellowship

## INVITED TALKS & ACADEMIC SERVICES

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### Google Speech Technologies Group

July 2025

*Invited Talk*

Google DeepMind

- Presented our "Multilingual and Multi-Accent Jailbreaking of Audio LLMs" paper to the NFM Reading Group led by the Speech Technologies Group at Google DeepMind.

### Program Committee Member (Reviewer)

- **2026:** ICLR, ICASSP, ICML (**Best Reviewer**), COLM, NeurIPS
- **2025:** ICLR, COLM XLLM-Workshop

## WORK EXPERIENCE

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### Dolby Laboratories

June 2026 – September 2026

*Research Intern, Supervisor: Deepak Chandran, Mei Chen*

San Francisco, California

- Working on interactive audio model safety

### Brave Software

June 2025 – September 2025

*Research Intern, Supervisor: Ali Shahin Shamsabadi*

London, United Kingdom (Remote)

- Worked on privacy & security of AI agents

### Super Chain AI (Conard International)

June 2021 – August 2021

*NLP Engineer Intern, Supervisor: Cat Yung*

Kowloon Bay, Hong Kong

- In charge of topic modeling and semantic analysis based on customer reviews and assigning specific semantics to the topics extracted.
- Competitors' analysis through web-scraping customer reviews from other drop-shipping websites.

### Military Service at Head Quarter of 12th Infantry Division

July 2018 – March 2020

*Sergeant of Republic of Korea Army*

Injae, Kang Won Do, Republic of Korea

- Officer Administrative Clerk Specialist
- Squad Leader of the Head Quarter

## SKILLS / LANGUAGES

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**Programming Language:** Python, C++

**Languages:** Korean (Native), English (Native), Chinese (Fluent)